

Assignment No 1 B(Classification)

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Subject : ADS Lab

Problem Statement: Assignment based on implementation of Artificial Neural Network (ANN) using TensorFlow/Keras for prediction problems, Exemplars: Student Performance Prediction, House Price Prediction, Employee Attrition Prediction

Implementation:

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

df = pd.read_csv("data.csv")

print(df.head())
print(df.info())

# -----
# Remove unnecessary columns
# -----
df.drop(["id"], axis=1, inplace=True)

if "Unnamed: 32" in df.columns:
    df.drop(["Unnamed: 32"], axis=1, inplace=True)

encoder = LabelEncoder()
df["diagnosis"] = encoder.fit_transform(df["diagnosis"])

# -----
# Features and Target
# -----
X = df.drop("diagnosis", axis=1)
y = df["diagnosis"]
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# -----
# Feature Scaling
# -----
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# -----
# Build ANN Model
# -----
model = Sequential()

model.add(Dense(32, activation='relu', input_shape=(X_train.shape[1],)))
model.add(Dense(16, activation='relu'))
model.add(Dense(1, activation='sigmoid'))

# -----
# Compile Model
# -----
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# -----
# Train Model
# -----
history = model.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=16,
    validation_split=0.2,
    verbose=1
)

```

```

# -----
# Evaluate Model
# -----
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Loss:", loss)
print("Test Accuracy:", accuracy)

# -----
# Prediction
# -----
y_pred = model.predict(X_test)

# Convert probability to binary
y_pred = (y_pred > 0.5)

print("\nActual Values:")
print(y_test.values[:10])

print("\nPredicted Values:")
print(y_pred[:10].flatten())

# -----
# Plot Accuracy
# -----
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')

plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("ANN Accuracy")
plt.legend()
plt.show()

```

Output:

```

      id diagnosis  ... fractal_dimension_worst  Unnamed: 32
0  842302      M  ...             0.11890         NaN
1  842517      M  ...             0.08902         NaN
2  84300903     M  ...             0.08758         NaN
3  84348301      M  ...             0.17300         NaN
4  84358402      M  ...             0.07678         NaN

```

```

[5 rows x 33 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568

```

Data columns (total 33 columns):

#	Column	Non-Null Count	Dtype
0	id	569 non-null	int64
1	diagnosis	569 non-null	object
2	radius_mean	569 non-null	float64
3	texture_mean	569 non-null	float64
4	perimeter_mean	569 non-null	float64
5	area_mean	569 non-null	float64
6	smoothness_mean	569 non-null	float64
7	compactness_mean	569 non-null	float64
8	concavity_mean	569 non-null	float64
9	concave points_mean	569 non-null	float64
10	symmetry_mean	569 non-null	float64
11	fractal_dimension_mean	569 non-null	float64
12	radius_se	569 non-null	float64
13	texture_se	569 non-null	float64
14	perimeter_se	569 non-null	float64
15	area_se	569 non-null	float64
16	smoothness_se	569 non-null	float64
17	compactness_se	569 non-null	float64
18	concavity_se	569 non-null	float64
19	concave points_se	569 non-null	float64
20	symmetry_se	569 non-null	float64
21	fractal_dimension_se	569 non-null	float64
22	radius_worst	569 non-null	float64
23	texture_worst	569 non-null	float64
24	perimeter_worst	569 non-null	float64
25	area_worst	569 non-null	float64
26	smoothness_worst	569 non-null	float64
27	compactness_worst	569 non-null	float64
28	concavity_worst	569 non-null	float64
29	concave points_worst	569 non-null	float64
30	symmetry_worst	569 non-null	float64
31	fractal_dimension_worst	569 non-null	float64
32	Unnamed: 32	0 non-null	float64

dtypes: float64(31), int64(1), object(1)

memory usage: 146.8+ KB

None

Epoch 1/50

1/23 ————— 39s 2s/step - accuracy:

0.2500 - loss: 0.821/23 ————— 0s

3ms/step - accuracy: 0.7083 - loss: 0.523/23

————— 2s 19ms/step - accuracy: 0.7253 -

loss: 0.5856 - val_accuracy: 0.8791 - val_loss: 0.4452

Epoch 2/50

1/23 ————— 1s 49ms/step - accuracy:
0.8750 - loss: 0.22/23 ————— 0s 2ms/step
- accuracy: 0.9261 - loss: 0.323/23 —————
0s 8ms/step - accuracy: 0.9258 - loss: 0.3539 - val_accuracy: 0.9231 - val_loss:
0.2956
Epoch 3/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9560 - loss: 0.223/23 —————
0s 8ms/step - accuracy: 0.9560 - loss: 0.2299 - val_accuracy: 0.9231 - val_loss:
0.2179
Epoch 4/50
1/23 ————— 0s 44ms/step - accuracy:
0.9375 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9698 - loss: 0.123/23 —————
0s 8ms/step - accuracy: 0.9698 - loss: 0.1629 - val_accuracy: 0.9451 - val_loss:
0.1732
Epoch 5/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9753 - loss: 0.123/23 —————
0s 8ms/step - accuracy: 0.9753 - loss: 0.1248 - val_accuracy: 0.9451 - val_loss:
0.1458
Epoch 6/50
1/23 ————— 0s 43ms/step - accuracy:
0.9375 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9780 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9780 - loss: 0.0996 - val_accuracy: 0.9560 - val_loss:
0.1312
Epoch 7/50
1/23 ————— 1s 51ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 8ms/step
- accuracy: 0.9808 - loss: 0.0847 - val_accuracy: 0.9560 - val_loss: 0.1206
Epoch 8/50
1/23 ————— 0s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9780 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9780 - loss: 0.0744 - val_accuracy: 0.9670 - val_loss:
0.1130
Epoch 9/50
1/23 ————— 0s 42ms/step - accuracy:
0.9375 - loss: 0.22/23 ————— 0s 2ms/step
- accuracy: 0.9801 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9808 - loss: 0.0661 - val_accuracy: 0.9451 - val_loss:
0.1092
Epoch 10/50

1/23 ————— 1s 52ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9835 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9835 - loss: 0.0600 - val_accuracy: 0.9560 - val_loss:
0.1069
Epoch 11/50
1/23 ————— 0s 41ms/step - accuracy:
0.9375 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9835 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9835 - loss: 0.0553 - val_accuracy: 0.9560 - val_loss:
0.1031
Epoch 12/50
1/23 ————— 1s 51ms/step - accuracy:
0.9375 - loss: 0.22/23 ————— 0s 2ms/step
- accuracy: 0.9830 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9835 - loss: 0.0510 - val_accuracy: 0.9560 - val_loss:
0.1010
Epoch 13/50
1/23 ————— 1s 51ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9863 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9863 - loss: 0.0476 - val_accuracy: 0.9670 - val_loss:
0.0993
Epoch 14/50
1/23 ————— 0s 41ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 0.9863 - loss: 0.0445 - val_accuracy: 0.9560 - val_loss: 0.0977
Epoch 15/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9863 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9863 - loss: 0.0423 - val_accuracy: 0.9560 - val_loss:
0.0984
Epoch 16/50
1/23 ————— 1s 55ms/step - accuracy:
1.0000 - loss: 0.21/23 ————— 0s 3ms/step
- accuracy: 0.9851 - loss: 0.023/23 —————
0s 9ms/step - accuracy: 0.9863 - loss: 0.0398 - val_accuracy: 0.9451 - val_loss:
0.0959
Epoch 17/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9890 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9890 - loss: 0.0370 - val_accuracy: 0.9560 - val_loss:
0.0968
Epoch 18/50

1/23 ————— 1s 49ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9890 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9890 - loss: 0.0353 - val_accuracy: 0.9560 - val_loss:
0.0951
Epoch 19/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9918 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9918 - loss: 0.0330 - val_accuracy: 0.9560 - val_loss:
0.0960
Epoch 20/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 0.9918 - loss: 0.0316 - val_accuracy: 0.9560 - val_loss: 0.0960
Epoch 21/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 0.9918 - loss: 0.0297 - val_accuracy: 0.9451 - val_loss: 0.0973
Epoch 22/50
1/23 ————— 1s 47ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9918 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9918 - loss: 0.0281 - val_accuracy: 0.9451 - val_loss:
0.0978
Epoch 23/50
1/23 ————— 0s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0267 - val_accuracy: 0.9451 - val_loss:
0.0995
Epoch 24/50
1/23 ————— 0s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9918 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9918 - loss: 0.0256 - val_accuracy: 0.9451 - val_loss:
0.1000
Epoch 25/50
1/23 ————— 1s 48ms/step - accuracy:
1.0000 - loss: 0.21/23 ————— 0s 3ms/step
- accuracy: 0.9940 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9945 - loss: 0.0237 - val_accuracy: 0.9451 - val_loss:
0.1011
Epoch 26/50
1/23 ————— 0s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step

- accuracy: 0.9918 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9918 - loss: 0.0226 - val_accuracy: 0.9451 - val_loss:
0.1003
Epoch 27/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0216 - val_accuracy: 0.9451 - val_loss:
0.1013
Epoch 28/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0203 - val_accuracy: 0.9451 - val_loss:
0.1039
Epoch 29/50
1/23 ————— 0s 41ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 0.9945 - loss: 0.0190 - val_accuracy: 0.9451 - val_loss: 0.1059
Epoch 30/50
1/23 ————— 1s 47ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 0.9945 - loss: 0.0182 - val_accuracy: 0.9451 - val_loss: 0.1059
Epoch 31/50
1/23 ————— 0s 41ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 10ms/step - accuracy: 0.9945 - loss: 0.0174 - val_accuracy: 0.9451 - val_loss:
0.1071
Epoch 32/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.22/23 ————— 0s 2ms/step
- accuracy: 0.9943 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0162 - val_accuracy: 0.9451 - val_loss:
0.1102
Epoch 33/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0156 - val_accuracy: 0.9451 - val_loss:
0.1110
Epoch 34/50
1/23 ————— 0s 43ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————

0s 8ms/step - accuracy: 0.9945 - loss: 0.0150 - val_accuracy: 0.9451 - val_loss:
0.1127
Epoch 35/50
1/23 ————— 1s 49ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 0.9945 - loss: 0.0137 - val_accuracy: 0.9451 - val_loss:
0.1150
Epoch 36/50
1/23 ————— 3s 173ms/step - accuracy:
0.9375 - loss: 0.23/23 ————— 0s 7ms/step -
accuracy: 0.9945 - loss: 0.0133 - val_accuracy: 0.9451 - val_loss: 0.1173
Epoch 37/50
1/23 ————— 0s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9945 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9945 - loss: 0.0123 - val_accuracy: 0.9451 - val_loss:
0.1175
Epoch 38/50
1/23 ————— 1s 51ms/step - accuracy:
1.0000 - loss: 3.23/23 ————— 0s 2ms/step
- accuracy: 0.9973 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9973 - loss: 0.0115 - val_accuracy: 0.9451 - val_loss:
0.1205
Epoch 39/50
1/23 ————— 0s 42ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 8ms/step
- accuracy: 0.9973 - loss: 0.0113 - val_accuracy: 0.9451 - val_loss: 0.1221
Epoch 40/50
1/23 ————— 1s 48ms/step - accuracy:
1.0000 - loss: 1.23/23 ————— 0s 7ms/step
- accuracy: 0.9973 - loss: 0.0103 - val_accuracy: 0.9451 - val_loss: 0.1243
Epoch 41/50
1/23 ————— 1s 47ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9973 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9973 - loss: 0.0099 - val_accuracy: 0.9451 - val_loss:
0.1251
Epoch 42/50
1/23 ————— 1s 48ms/step - accuracy:
1.0000 - loss: 9.23/23 ————— 0s 2ms/step
- accuracy: 0.9973 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9973 - loss: 0.0091 - val_accuracy: 0.9451 - val_loss:
0.1268
Epoch 43/50

1/23 ————— 2s 91ms/step - accuracy:
1.0000 - loss: 0.22/23 ————— 0s 2ms/step
- accuracy: 1.0000 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 1.0000 - loss: 0.0086 - val_accuracy: 0.9451 - val_loss:
0.1277
Epoch 44/50
1/23 ————— 0s 44ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 0.9973 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 0.9973 - loss: 0.0081 - val_accuracy: 0.9451 - val_loss:
0.1305
Epoch 45/50
1/23 ————— 1s 51ms/step - accuracy:
1.0000 - loss: 0.20/23 ————— 0s 3ms/step
- accuracy: 1.0000 - loss: 0.023/23 —————
0s 8ms/step - accuracy: 1.0000 - loss: 0.0079 - val_accuracy: 0.9451 - val_loss:
0.1324
Epoch 46/50
1/23 ————— 1s 47ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 1.0000 - loss: 0.0074 - val_accuracy: 0.9451 - val_loss: 0.1373
Epoch 47/50
1/23 ————— 0s 40ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 7ms/step
- accuracy: 1.0000 - loss: 0.0069 - val_accuracy: 0.9451 - val_loss: 0.1350
Epoch 48/50
1/23 ————— 4s 184ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step -
accuracy: 1.0000 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 1.0000 - loss: 0.0064 - val_accuracy: 0.9451 - val_loss:
0.1361
Epoch 49/50
1/23 ————— 1s 45ms/step - accuracy:
1.0000 - loss: 0.23/23 ————— 0s 2ms/step
- accuracy: 1.0000 - loss: 0.023/23 —————
0s 7ms/step - accuracy: 1.0000 - loss: 0.0062 - val_accuracy: 0.9451 - val_loss:
0.1406
Epoch 50/50
1/23 ————— 0s 44ms/step - accuracy:
1.0000 - loss: 9.23/23 ————— 0s 7ms/step
- accuracy: 1.0000 - loss: 0.0062 - val_accuracy: 0.9451 - val_loss: 0.1432
1/4 ————— 0s 33ms/step - accuracy:
0.9688 - loss: 0.194/4 ————— 0s
12ms/step - accuracy: 0.9737 - loss: 0.0904
Test Loss: 0.09035109728574753
Test Accuracy: 0.9736841917037964

4/4 ————— 0s 27ms/step

Actual Values:

[0 1 1 0 0 1 1 1 0 0]

Predicted Values:

[False True True False False True True True False False]

